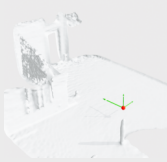
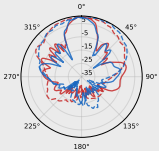


Differentiable Parameters

Radar Parameters

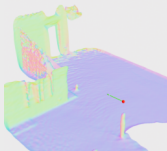


Sensor Pose

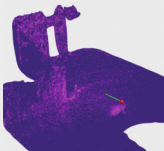


Beam Patterns

Scene Parameters



Positions & Normals



Materials

Forward Model

MIMO Ray Tracing

$$\hat{p}_r^{(D)} = \frac{P_t \lambda^2}{(4\pi)^2 N} \sum_{k=1}^N \frac{G_{r,k}}{\rho_{rx,k}} \prod_{\ell=1}^{D-1} \frac{f_{\ell} G_{\ell}}{\rho_{\ell}} \cdot \frac{G_{t,k} c_{D \rightarrow tx} f_D}{d_{D,tx}^2} V_k$$

(§3.1)



mmWave BSDF

$$f = A[\eta f_{\text{coh}} + (1 - \eta) f_{\text{inc}}]$$

(§3.2)



Ray Generation

Reservoir · Specular Manifold · Diffraction

(§3.3)



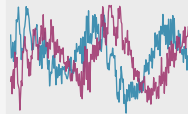
MIMO Coherent ADC

$$s[k] = \sum_p a_p e^{j2\pi(f_c + Sk)\tau_p}$$

(§3.1)

Rendered Prediction

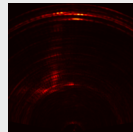
Rendered ADC



Complex-valued ADC



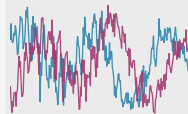
Rendered RA



Magnitude-only RA

Ground Truth

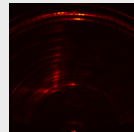
GT ADC



Complex-valued ADC



GT RA



Magnitude-only RA

RA Loss

